Internship Progress Report

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Topic: Unsupervised Domain Adaptation for Hyperspectral Mangrove Mapping.

1.Data Description:

Two hyperspectral datasets from the Krishna mangrove region were used:

- Source Dataset: 2019 hyperspectral image
- Target Dataset: 2024 hyperspectral image
- Number of spectral bands: 64

2.Methodology:

2.1 Data Preprocessing:

The following preprocessing steps were applied:

- Resampling and spatial alignment of hyperspectral datasets.
- Robust feature scaling applied independently to source and target datasets.
- Extraction of stable pixels using cosine-distance thresholding.
- Computation of covariance statistics and subspace representations.

2.2 Stable Pixel Selection:

Stable pixels were identified using cosine distance-based similarity analysis.

- Stable pixel percentile threshold: 2%
- Stable sample count: 41,937 pixels

These stable pixels were used for estimating domain adaptation transformations.

2.3 Domain Adaptation Techniques:

When source and target domain follows different statistical distributions then classical ML fails to work and doesn't give perfect fitting of trained model for testing. To overcome this limitation we try to super impose source statistical distribution onto Target statistical distribution which creates similarity between two distributions, This process is called Domain Adaptation.

A. **CORAL (Correlation Alignment):** CORAL aligns second-order statistics between source and target domains by matching covariance matrices.

B. **Subspace Alignment (SA):** SA aligns source and target PCA subspaces through a linear transformation.

C. **CORAL followed by SA:** A sequential alignment strategy was evaluated in which CORAL was first applied to perform covariance matching, followed by Subspace Alignment to further reduce residual geometric differences between source and target domains.

D. **SA followed by CORAL**: A second sequential strategy was evaluated in which SA was first applied to reduce subspace discrepancy, followed by CORAL to align covariance statistics.

E. **Transfer Component Analysis (TCA):** TCA learns a common latent feature space for source and target domains by minimizing Maximum Mean Discrepancy (MMD).

2.4 Evaluation Metrics:

Two quantitative metrics were used:

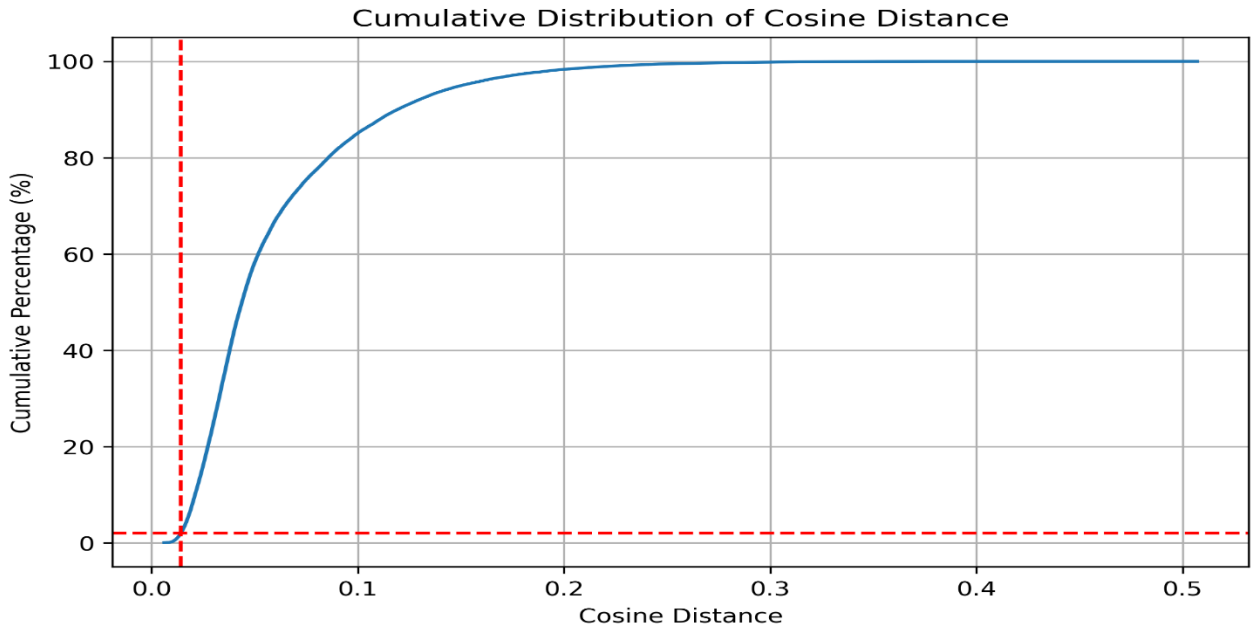
A. **Covariance Gap:** The Frobenius norm between source and target covariance matrices, Covariance Gap = ||Cs − Ct||F

Lower values indicate better statistical alignment.

B. **Subspace Distance:** Distance between source and target PCA subspaces, Subspace Distance = ||Ps − Pt||F
Lower values indicate stronger geometric alignment between domains.

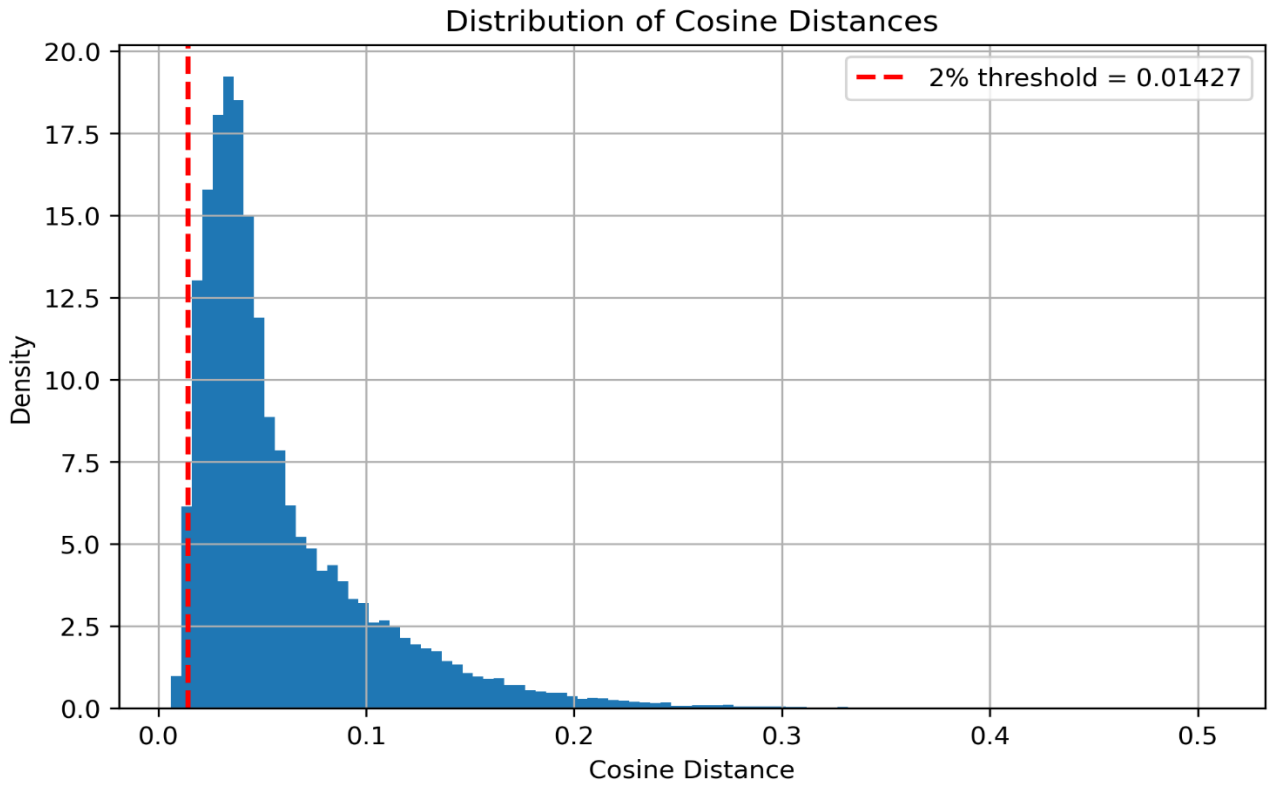
3.Expiremental Results:

3.1 Stable Pixel Selection:



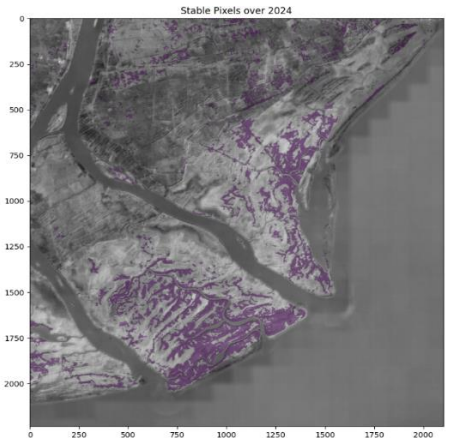
Cosine similarity = 1 – Cosine distance.

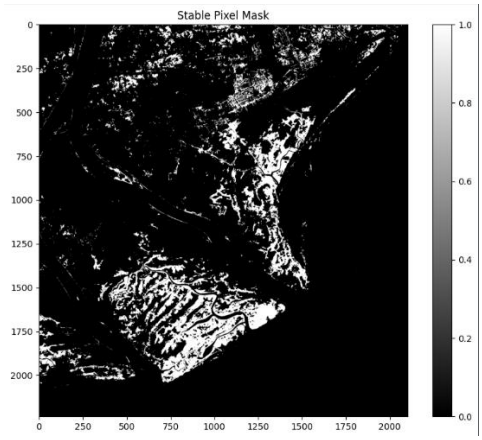
By the above plot of Cumulative distribution of Cosine distance, cosine distance is very small for lower percentiles and gradually increases rapidly to certain extent showing of inclusion of stable pixels and from that curve increases slowly and tend to constant showing that inclusion of unstable pixels starts from that point.



We can see that stable pixel density is maximum at 5% threshold and reduces from that showing inclusion of unstable pixels start from 5% threshold level.

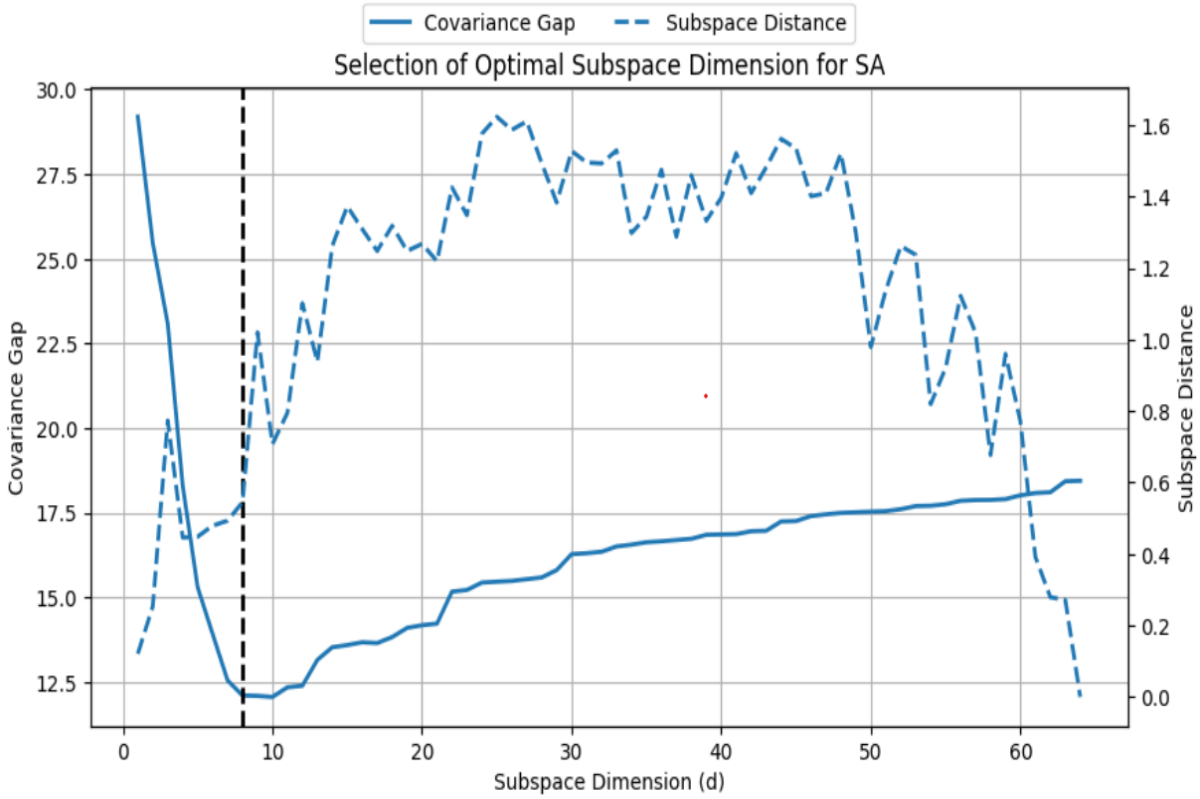
Highest similarity is shown at 1% threshold level and maximum number of stable pixels are obtained around 5% threshold level. We cannot choose 1% percent threshold level as the stable pixels are insufficient and also we cannot choose 5% percent threshold level as similarity is very less. Hence, we choose 2% threshold level which contains sufficient number of stable pixels and next best similarity level after 1% threshold level.





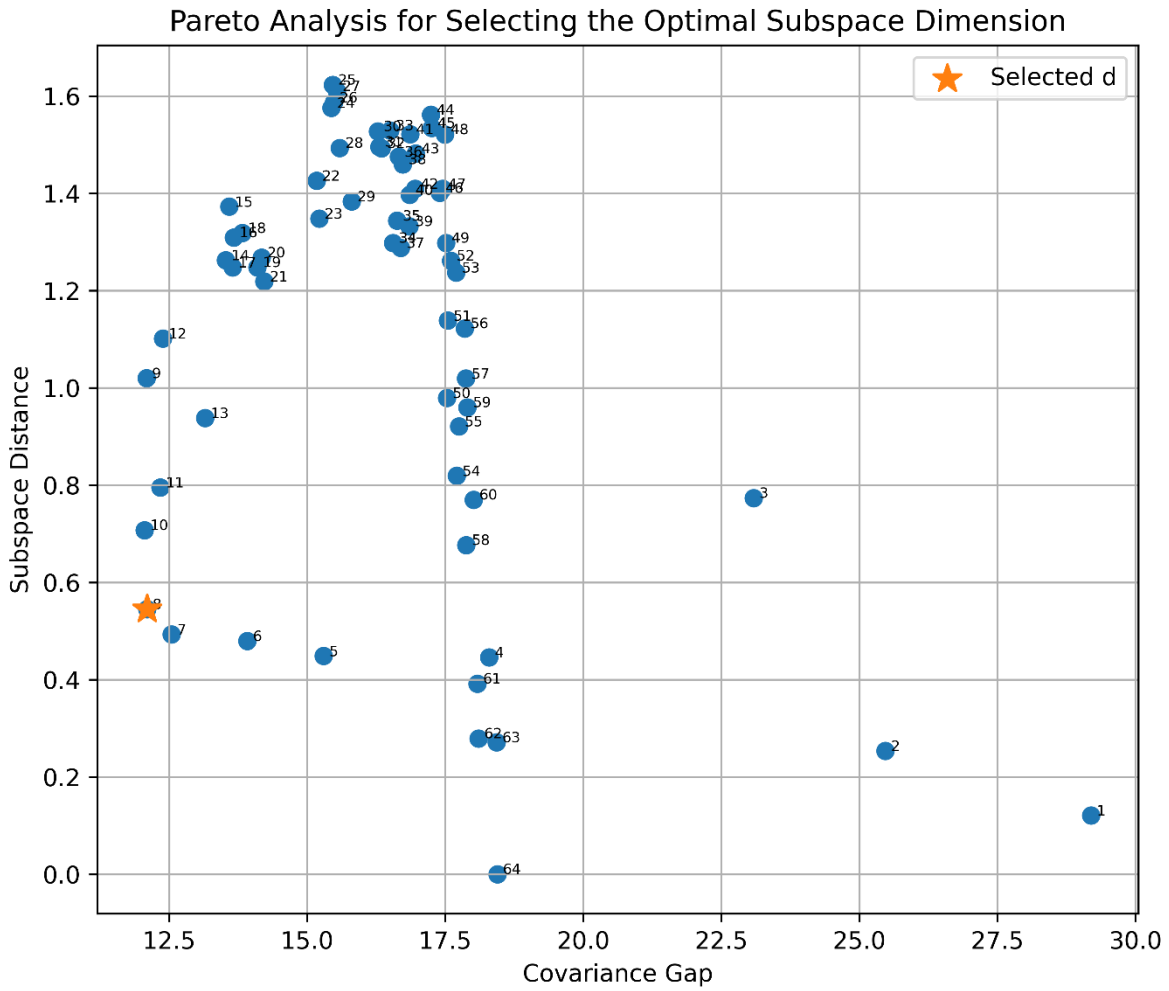
3.2 SA Subspace Dimension Selection:

Subspace Alignment (SA) requires the selection of an appropriate subspace dimension ((d)), which determines the number of principal components used to represent the source and target domains. Selecting a very small value of (d) may result in the loss of important spectral information, whereas a very large value may retain redundant information and reduce the effectiveness of domain adaptation. Therefore, an experimental analysis was conducted by varying the subspace dimension from 1 to 64 principal components.



For each value of (d), the aligned source and target domains were evaluated using two complementary metrics: the covariance gap, which quantifies the difference in second-order statistical properties, and the subspace distance, which measures the geometric similarity between the aligned feature spaces. Above figure illustrates the variation of these two metrics with increasing subspace dimension. The results indicate that increasing the subspace dimension progressively improves the geometric alignment between the source and target domains, as evidenced by the reduction in subspace distance. However, the covariance gap exhibits a different trend, indicating that increasing the number of principal components does not necessarily lead to improved statistical alignment. Beyond a certain dimensionality, the reduction in covariance discrepancy becomes marginal while additional components contribute little to further alignment.

Although smaller dimensions ((d = 1)–(4)) achieve lower subspace distances, they exhibit considerably larger covariance gaps, indicating insufficient statistical alignment between the source and target domains. Conversely, larger dimensions improve one metric at the expense of the other and do not provide a balanced alignment.



The selected dimension, **(d = 8)**, represents an effective compromise between statistical and geometric alignment. At this point, both the covariance gap and subspace distance are simultaneously low compared with neighbouring dimensions, placing it close to the Pareto-optimal region. Therefore, (d = 8) was selected as the optimal subspace dimension for the Subspace Alignment experiments, providing a balanced representation that preserves discriminative spectral information while effectively reducing domain discrepancy.

Based on the combined behaviour of both evaluation metrics, the selected subspace dimension represents a balance between preserving discriminative spectral information and achieving effective domain alignment. This dimension was subsequently used for all Subspace Alignment experiments presented in the following sections.

3.3 Domain Adaptation:

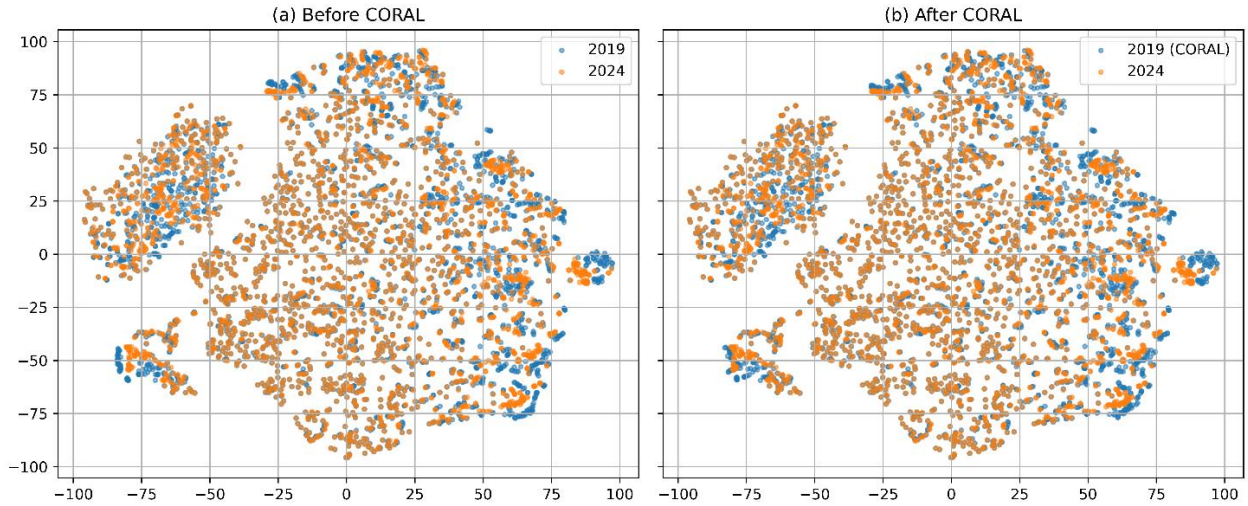
Initial Covariance gap and Subspace distance between 2019 and 2024 datasets of Krishna delta region are 18.44 and 6.18 respectively. We need to reduce these two parameters as low as possible to obtain the best superimposition and getting a best fit for classifier model for further proceeding to classify mangrove types based on regional domain transfer.

Method	Covariance Gap (Before)	Covariance Gap (After)	Subspace Distance (Before)	Subspace Distance (After)
Original (No Adaptation)	18.443	18.443	6.184	6.184
Subspace Alignment (SA)	18.443	12.100	6.184	0.545
Transfer Component Analysis (TCA)	18.525	6.182	6.184	0.869
SA + CORAL	18.443	5.415	6.184	0.545
CORAL + SA	18.443	3.902	6.184	0.00017
CORAL	18.443	0.00175	6.184	0.00027

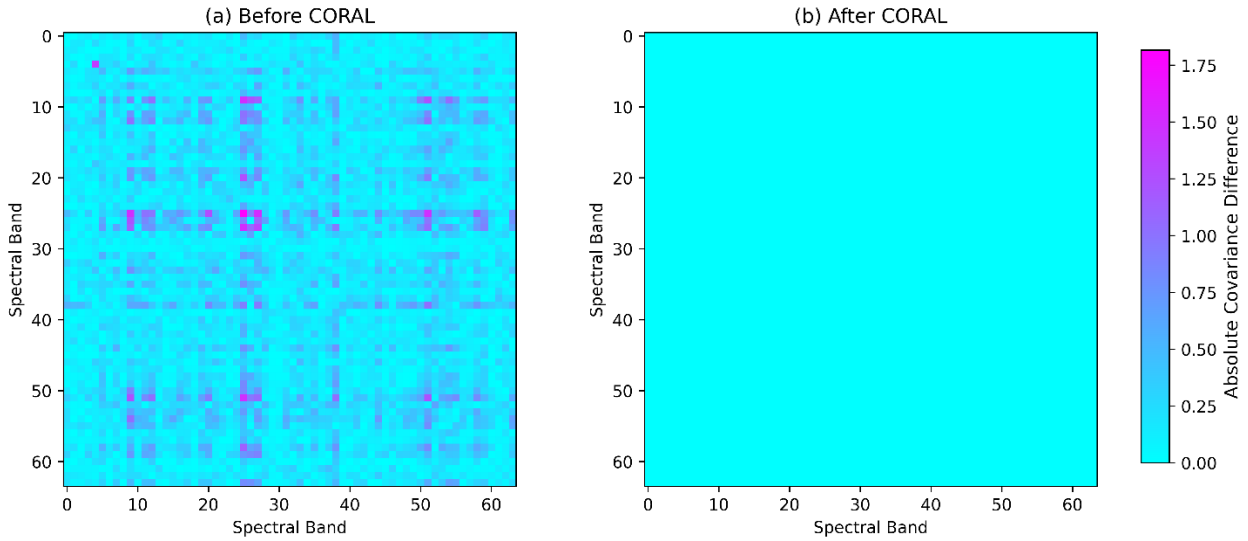
The quantitative comparison demonstrates that CORAL consistently achieved the most effective domain adaptation among all evaluated shallow methods. While SA substantially reduced the geometric discrepancy between source and target subspaces, its covariance alignment remained limited. TCA produced a greater reduction in covariance discrepancy than SA but did not achieve comparable subspace alignment. Hybrid approaches (SA+CORAL and CORAL+SA) improved one alignment criterion at the expense of the other, indicating that sequential transformations did not outperform a single CORAL transformation. Consequently, CORAL was selected as the final shallow domain adaptation method for subsequent cross-region mangrove species classification experiments due to its superior performance in simultaneously minimizing both covariance gap and subspace distance.

3.4 Qualitative Vizualizations:

To qualitatively assess the effectiveness of the selected domain adaptation method, the stable pixel samples were projected into a two-dimensional feature space using dimensionality reduction. Below figure presents the feature distributions before and after CORAL-based domain adaptation by t-distributed Stochastic Neighbour Embedding (t-SNE) was employed. Prior to applying t-SNE, the 64-band spectral features were reduced to 20 principal components using Principal Component Analysis (PCA) to suppress noise and improve computational efficiency. The reduced features were then embedded into a two dimensional space using a single t-SNE model, ensuring that the source, target, and CORAL-aligned samples were represented within the same embedding space.



The t-SNE visualization demonstrates that the source (2019) and target (2024) datasets exhibit a high degree of overlap even before domain adaptation, indicating that they share similar local feature structures. However, several localized regions show noticeable differences in sample distribution, reflecting the presence of domain shift. After applying CORAL, the transformed source samples display improved overlap and mixing with the target samples while preserving the overall structure of the feature space. Although the visual changes are subtle, the reduction in localized discrepancies indicates that CORAL has successfully aligned the source distribution closer to the target distribution without distorting the intrinsic spectral relationships. These qualitative observations are consistent with the substantial reductions observed in the covariance gap and subspace distance, confirming the effectiveness of CORAL for domain adaptation.



The covariance difference matrices provide a visual comparison of the statistical discrepancy between the source and target domains before and after CORAL-based domain adaptation. Figure (a) exhibits several high-intensity regions, indicating substantial differences in the covariance structure of the 2019 and 2024 hyperspectral datasets across multiple spectral bands. These regions correspond to large second-order statistical discrepancies that contribute to the observed domain shift.

Following CORAL transformation, the covariance difference matrix in Figure (b) becomes predominantly low in intensity, with only a few localized regions exhibiting noticeable differences. This reduction in high-intensity areas indicates that the covariance structure of the transformed source data closely matches that of the target domain. The visual reduction in covariance discrepancy is consistent with the quantitative results, where the covariance gap decreased from **18.44** before adaptation to **0.00175** after CORAL. These results demonstrate that CORAL effectively aligns the second-order statistical properties of the source and target domains, thereby reducing the domain shift while preserving the underlying spectral relationships.

4.Conclusion:

The experimental evaluation demonstrates that CORAL is the most effective shallow domain adaptation technique among the methods investigated. Compared with SA and TCA, CORAL produced the lowest covariance gap and subspace distance, indicating strong statistical and geometric alignment between the source and target hyperspectral domains.

Based on these results, CORAL has been selected as the primary domain adaptation method for subsequent cross-region mangrove species classification experiments.

5.Future Work:

The next phase of the research will focus on:

A. Training a mangrove species classifier using labeled samples from the Krishna mangrove region.
B. Applying CORAL-based domain adaptation to transfer the learned feature space to:
    - Malad Mangroves (Maharashtra)
    - Pichavaram Mangroves (Tamil Nadu)
C. Evaluating species-level classification performance across regions.
D. Assessing the effectiveness of shallow domain adaptation for cross-region mangrove species mapping.